Investigating with commands

NUnix · L3 MIAGE · CM2 · 90 minutes

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Investigating with commands

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A computer is slow. Would adding RAM help?
Check which resource is under pressure first.

We will use commands to inspect the system, find useful information, and turn it into a summary we can save.

CM2 outline

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1. Processes and resources
2. Finding files and text
3. Inputs, outputs and redirections
4. Combining commands with pipes

Observe the system → find information → redirect streams → combine commands.

1. Processes and resources

Which programs are running, and where are the resources going?

CPU, memory and disk space

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Resource	Question	First observation
CPU	Is computation keeping a processor busy?	`top`, `ps`
RAM	Is memory pressure affecting the system?	`free -h`
Disk	Can the filesystem store more data?	`df -h .`

Adding RAM does not create free space on a full disk.

Running in the foreground

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A **process** is a running instance of a program.

```
sleep 10
```

The shell waits for this foreground command to finish before showing a new prompt.

Ctrl+C requests an interrupt of the foreground job.

Inspecting processes

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```
ps
ps -ef
```

Field	Meaning
`PID`	Process identifier
`TTY`	Associated terminal, if any
`TIME`	Accumulated CPU time
`CMD`	Command

Checking memory usage

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```
free -h
ps -eo pid,comm,%cpu,%mem --sort=-%mem
```

- Read **available** memory; cache use is normal.
- Identify the application’s process and follow it over time.
- Check logs before attributing every delay to memory.

Checking disk usage

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Tool	Scope	Question
`df`	A filesystem	How much space is available?
`du`	A directory tree	Which directories use space?
`find`	Individual entries	Which files fit my criteria?

Start broad, then inspect a **specific file**.

Find which directories use space

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```
df -h "$HOME"
du -h --max-depth=1 "$HOME"
```

`df` shows usage of the filesystem containing your home.
`du` shows how much space your directories use.

From resources to files

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1. Processes and resources
2. **Finding files and text**
3. Inputs, outputs and redirections
4. Combining commands with pipes

We can inspect CPU, memory and disk usage. Now we need to locate files and find the relevant lines in their contents.

2. Finding files and text

Where is the file, and which lines contain the information we need?

Finding files

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```
find . -type f -name '*.txt'
```

- `.` is the **starting directory**.
- `find` also visits its subdirectories.
- `-type f` selects regular files.
- `-name '*.txt'` tests each entry's name.

The quotes let **find** receive the pattern unchanged.

Search criteria

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Criterion	Matches
`-iname '*.txt'`	Name, ignoring case
`-user alice`	Entries owned by Alice
`-type d` / `-type l`	Directories / symbolic links
`-size +100M`	Size greater than 100 MiB
`-mtime -2`	Modified less than 48 hours ago

```
find . -type f -size +100M
```

Checkpoint: find

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1. Find regular `.c` files below the current directory.
2. Why quote `'*.c'`?

1. `find . -type f -name '*.c'`
2. So the shell passes the pattern to `find` without expanding it first.

A small log file

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We have located a log file. Next, we will search its contents with `grep`. Save these four lines as `app.log`:

```
10:00 INFO STARTED
10:01 ERROR DISK_FULL
10:02 ERROR TIMEOUT
10:03 ERROR DISK_FULL
```

Field	Meaning
1	Time
2	Severity
3	Event code

Finding text with grep

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```
grep ' ERROR ' app.log
grep -r ' ERROR ' .
grep -n 'DISK_FULL' app.log
```

- Print **matching lines** from a file.
- `-r` searches directories recursively.
- `-n` includes line numbers.

Exit status: **0** = match, **1** = no match, **2** = error in GNU `grep`.

Where the name grep comes from

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The editor ed had the command pattern:

```
g/re/p
```

- **g**: apply globally to matching lines.
- **re**: a regular expression.
- **p**: print those lines.

`grep` makes that text search available as a separate command.

Patterns in grep

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Regular expression	Meaning
`.`	Any one character
`m*`	Zero or more repetitions of `m`
`[1-3]`	A digit from 1 to 3

```
grep 'm.*t' names.txt
grep 'toto[1-3]' names.txt
```

In a regular expression, * repeats the **preceding item**.

Who interprets this pattern?

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```
printf '%s\n' *.log
grep 'DISK.*' app.log
find . -name '*.log'
```

1. The **shell** expands a filename pattern.
2. **grep** interprets a regular expression.
3. **find** interprets a quoted filename pattern.

From matching lines to saved results

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1. Processes and resources
2. Finding files and text
3. **Inputs, outputs and redirections**
4. Combining commands with pipes

We can find files and select useful lines. How can a command read from a file and save its output or errors?

3. Inputs, outputs and redirections

Where does a command read, where does it write, and did it succeed?

Input, output and status

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Channel	Number	Usual terminal behaviour
`stdin`	**0**	Read keyboard input
`stdout`	**1**	Display normal output
`stderr`	**2**	Display diagnostic messages

The **exit status** is a number reported when a command finishes. It is separate from the text it prints.

Exit status

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```
true
echo "$?"
false
echo "$?"
```

0 means success / true. **Nonzero** means failure / false.
`$?` is the status of the **most recent command**.

Reading input from a file

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```
sort < names.txt
```

The shell opens `names.txt` as **standard input** for `sort`.

`sort` writes the sorted lines to standard output, which still points to the terminal.

Saving output

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```
sort names.txt > sorted.txt
echo 'Zoé' >> sorted.txt
```

Operator	Effect on the destination file
>	Create it, or **replace** its contents
>>	Create it, or **append** to its contents

Redirecting errors

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```
find . -name '*.txt' > matches.txt 2> errors.txt
```

- > redirects **stdout**, descriptor 1.
- 2> redirects **stderr**, descriptor 2.
- Normal results and diagnostics go to separate files.

Redirections are applied from left to right

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```
ls notes.txt missing > all.txt 2>&1
ls notes.txt missing 2>&1 > out.txt
```

Command	stdout ends up…	stderr ends up…
First	In `all.txt`	In `all.txt`
Second	In `out.txt`	At the terminal

2>&1 copies stdout's destination **at that moment**.
It does not make stderr follow later changes to stdout.

Why did the file become empty?

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```
sort names.txt > names.txt
```

The shell opens and truncates `names.txt` **before** starting `sort`. The input is already gone.

Write to a **different file**, then inspect the result.

From one command to a pipeline

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1. Processes and resources
2. Finding files and text
3. Inputs, outputs and redirections
4. **Combining commands with pipes**

We can choose the input and output of one command. Now we will connect commands to count and rank the errors in our log.

4. Combining commands with pipes

How can one command use another command’s output?

A pipe between two commands

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```
grep ' ERROR ' app.log | wc -l
```

`grep` writes matching lines to its standard output.
The pipe connects that output to `wc`'s standard input.

The result is the **number of matching lines**.

Connecting programs like garden hoses

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Doug McIlroy described connecting programs so that one program's output could become another's input.

```
grep ' ERROR ' app.log | wc -l
```

Each tool has a small job.
The shared text stream lets us combine them.

Pipeline step 1: extract error codes

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```
grep ' ERROR ' app.log |
    cut -d ' ' -f 3
```

```
DISK_FULL
TIMEOUT
DISK_FULL
```

Pipeline step 2: count each code

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```
grep ' ERROR ' app.log |
    cut -d ' ' -f 3 |
    sort | uniq -c | sort -nr
```

```
      2 DISK_FULL
      1 TIMEOUT
```

`uniq -c` counts **adjacent** equal lines. Sort first.

Save and display output with tee

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```
sort names.txt | tee sorted.txt
```

`tee` copies its input to a file **and** standard output.

Think of a T-shaped pipe fitting: the stream has two outlets.

Combining arguments and redirections

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```
sort < names.txt > sorted.txt
sort < names.txt | head -n 3 > first-three.txt
```

- `< names.txt` selects the input stream.
- `-n 3` remains an **argument** to head.
- `> first-three.txt` selects the output file.

Checkpoint: redirections

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1. Which operator appends to a file?
2. What does `a | b` connect?
3. Does `a > out.txt` also redirect standard error?

1. `>>`
2. a's stdout to b's stdin.
3. No. Use `2>` to redirect stderr.

The exit status of a pipeline

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```
false | sort
echo "$?"
set -o pipefail
false | sort
echo "$?"
```

Output: **0**, then **1**.
Bash normally reports the last pipeline command’s status.

CM2 summary: commands and pipelines

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1. **Processes and resources:** inspect CPU, memory and disk usage.
2. **Finding files and text:** locate files with `find`, select lines with `grep`.
3. **Inputs, outputs and redirections:** read files, save results, separate errors.
4. **Combining commands with pipes:** extract, count and rank log entries.

Optional reference

More search expressions and interactive process control

Combining criteria

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```
find /usr -name '*.c'
find . -user alice -name '*.c'
find . -not \( -name '*.obj' -o -name '*.map' \)
```

- Adjacent tests use **AND**. `-a` makes it explicit.
- `-o` means **OR**. `-not` negates a test.
- Escaped parentheses group tests for `find`.

Running a command on matches

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```
find . -type f -name '*.txt' -exec wc -l {} \;
```

- `-exec` starts the command to run.
- `{}` stands for the current matching path.
- `\;` ends the command without ending the shell's command line.

Running in the background

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```
sleep 60 &
jobs
fg %1
```

- & lets the shell return to the prompt immediately.
- `jobs` lists jobs managed by this shell.
- `fg %1` brings job **1** to the foreground.

Suspending and resuming a job

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Action	Effect
Ctrl+Z	Suspend the foreground job
`bg %1`	Resume job 1 in the background
`fg %1`	Resume job 1 in the foreground

A **job number** belongs to a shell.
A **PID** identifies a process in the system.

Sequential and asynchronous execution

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```
who | sort; date
sleep 10 &
date
```

- ; runs the next command after the previous one finishes.
- & starts work asynchronously.
- The final `date` can run while `sleep` is still active.

Signals and end of input

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Action	Meaning
`kill PID`	Send `SIGTERM` to a process
Ctrl+C	Send `SIGINT` to the foreground process group
`Ctrl+\`	Send `SIGQUIT` to the foreground process group
Ctrl+Z	Send `SIGTSTP` to suspend the foreground group
Ctrl+D	Terminal end-of-input action